Bill of Materials

India Sourcing Guide — Components, Pricing, and Purchase Links

COMPONENT	QTY	UNIT PRICE (₹)	TOTAL (₹)	SOURCING
ESP32 DevKit v1 (30-pin) Dual-core 240 MHz, built-in WiFi+BT	1	₹280–350	₹315	Robu.in / Evelta / Amazon
nRF24L01+ PA/LNA module With external SMA antenna — mandatory	1	₹150–220	₹185	Robu.in / Robokits / Amazon
ILI9341 TFT 2.4" w/ XPT2046 320x240, SPI, touch controller onboard	1	₹350–450	₹400	Robu.in / Evelta / Amazon
Passive buzzer (3-pin) 3.3V compatible, NOT active buzzer	1	₹20–40	₹30	Robu.in / local electronics
100µF electrolytic cap Across nRF24 VCC/GND — CRITICAL	2	₹5–10	₹7	Any local electronics shop
TP4056 charge module USB-C variant preferred, with protection	1	₹40–70	₹55	Robu.in / Amazon
3.7V LiPo 1000mAh+ 1000–2000 mAh; TP4056 wired to BAT+/BAT-	1	₹200–350	₹275	Robu.in / Amazon / local
Slide switch (SPDT) In-line on TP4056 OUT+ rail	1	₹10–20	₹15	Any local electronics shop
7x9cm perfboard 2.54mm pitch green; all components mount here	1	₹30–50	₹40	Robu.in / local
Male/female headers For ESP32 mount, pin breakouts	1	₹30–60	₹45	Any local electronics shop
Jumper wires (assorted) M-M and M-F for prototype wiring	1	₹50–80	₹65	Robu.in / Amazon
USB-A to USB-C cable For TP4056 charging (if USB-C module)	1	₹80–120	₹100	Amazon / local
Heat shrink / tape Insulation for LiPo leads and joints	1	₹20–40	₹30	Local electronics shop
TOTAL BUILD COST (approx.)			**₹1562**	Range: ₹1265–₹1860

■ **CRITICAL — nRF24L01 Power: Always use the PA/LNA model with external antenna. Place a 100µF capacitor directly across VCC and GND on the nRF24 module. The nRF24 MUST be powered from the ESP32 3.3V rail — never directly from TP4056 OUT+ (4.2V at full charge will damage the module).**

■ BOM Notes: Prices verified on Robu.in and Amazon.in approximately mid-2025. Components marked 'local electronics shop' are available at any Bengaluru SP Road / Chickpet or Lamington Road (Mumbai) electronics market for significantly cheaper than online. Buy 2–3 nRF24L01 modules — they are the most fragile part and the most likely to be counterfeit when purchased cheaply.

Online Sourcing Guide — India

PLATFORM	BEST FOR	SHIPPING	NOTES
robu.in	ESP32, nRF24, TFT, TP4056, perfboard, headers	3–5 days	Best selection for ESP32 ecosystem components. Legit stock.
evelta.com	ESP32, TFT modules, dev boards	3–5 days	Good quality control. Slightly pricier but reliable.
amazon.in	LiPo batteries, USB cables, buzzer, capacitors	1–3 days	Fast delivery. Check seller ratings for nRF24 — many counterfeits.
robokits.co.in	nRF24L01, RF modules, sensors	4–7 days	Specialised RF modules. Good for nRF24 PA/LNA sourcing.
SP Road, Bengaluru	Capacitors, resistors, perfboard, switches, wire	Same day	Cheapest prices for passives. Bring a component list.
Lamington Rd, Mumbai	All components	Same day	Most comprehensive physical market for electronics in India.

Tools Required for Assembly

› Soldering iron (25–40W) with fine tip — mandatory

› Solder wire (60/40 or lead-free, 0.8mm)

› Multimeter — for continuity checks before power-on

› Helping hands / PCB holder — for soldering headers to perfboard

› Wire stripper and cutter

› Heat gun or lighter — for heat shrink on LiPo leads

› Arduino IDE 2.x installed with ESP32 board package (Espressif v2.x or v3.x)

This BOM reflects the v3.1 ENGINEERING HARDENED build. The v7.0 roadmap includes a custom PCB, hardware fuel gauge IC, USB-C native charging, and 3D-printed enclosure — component list will differ.